

MAINS MASTER PROGRAM (MMP) 2024

SCIENCE AND TECHNOLOGY-2

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2. SPACE

1) PSLV

- Practice Questions:

- Why is Polar Satellite Launch Vehicle (PSLV) considering the workhorse of ISRO? Discuss the capabilities and achievements of PSLV [15 marks, 250 words]

- The PSLV is the third-generation satellite launch vehicle of India. It is an expandable system and was the first Indian Launch Vehicle to be equipped with Liquid Stage.

- Where is PSLV used?

- It was developed to allow India to launch its Indian Remote Sensing (IRS) satellite into Sun synchronous orbit, a service that was, until the advent of the PSLV, commercially available only from Russia. PSLV can also launch small size satellites into Geostationary Transfer Orbit.

- **Capabilities:** Payload to SSPO (sun synchronous polar orbit): 1,860 Kg; Payload to Sub GTO: 1,425 Kg

- **Key features of PSLV Engines:** PSLV has four stages using solid and liquid propulsion alternatively.
- **Expansion of capabilities: Strap on Motors:** PSLV uses 6 solid rocket strap-on motors to augment the thrust provided by the first stage in its PSLV-G (1678 kg in SSPO) and PSLV-XL (1750 kg to SSPO) variants. PSLV-DL, PSLV-QL versions use 2 and 4 straps on motors respectively. PSLV-CA (1100 kg in LEO) uses no strap on motors.

» Key Significance and Achievements of PSLV

- **Reliability:** Only 2 failures in almost 3 decades of service and 60 launches.
- **Commercial use:** PSLV has launched **more than 350 foreign satellites** from 34 different countries so far.
- It has played significant role in various major ISRO missions (including Chandryaan-1, MOM, IRNSS system etc.)
- **India's Soft Power:** PSLV by providing the capability to launch satellites for foreign countries enhances India's soft power in the region.
- Many **learnings** from the development of PSLV has helped scientists develop several non-space applications like fire resistant tiles, better engines for missiles etc.

» Conclusion

- Today, ISRO is playing a key role of connecting continents and people via its satellite launch business model. From education to mobile telephony/ communications and navigation to business, the footprint of ISRO is becoming all pervasive.

2) GSLV MK-2

- **Practice Question:** Why has GSLV-Mk-2 acquired the nickname of **naughty boy**. Discuss its key launches of last one year. [10 marks, 150 words]
- **Introduction:**
 - » GSLV is the 4th generation launch vehicle of ISRO. It is an expandable launch system. It was primarily developed to launch INSAT class of communication satellite into Geosynchronous Transfer Orbit (GTO). It is currently being used for launching GSAT series
- **Nickname: Naughty Boy:** It has been given this nickname because of its rather patchy track record. So far (as of June 2024), GSLV has had 16 launches, but 25% of them (i.e. 4/16) have been failures.
 - » **Reason for failures:** The main problem is the Cryogenic engine that powers the third and final stage – the Cryogenic Upper Stage (CUS). This engine is a reverse engineered version of Russian Technology. Due to MTCR restrictions Russia couldn't provide India with the technology and India had to reverse engineer the Russian technology which had some flaws.
- **Technology and Capabilities:**
 - » It is a three-stage engine with first stage being solid, second liquid and third being cryogenic engine.
 - » **Payload to GTO:** Presently GSLV can inject 2.5 ton (GSLV Mk-2) of communication satellite into Geosynchronous Transfer Orbit and it can inject 5 tonnes of satellite in LEO.
- **Recent Launches:**
 - 1) **GSLV-F12/ NVS-01 Mission:** Accomplished successfully in May 2023. This GSLV mission deployed NVS-01 navigation satellite, weighing about 2232 kg, into GTO. The vehicle lifted off from the second launch pad at SDSC-SHAR, Sriharikota and injected the satellite after about 19 minutes of flight
 - 2) **GSLV F14/ INSAT-3DS Mission:** Accomplished in Feb 2024, it was the 16th mission of GSLV. It deployed the INSAT-3DS meteorological satellite in GTO. It was a mission fully funded by ministry of earth science.
- **Conclusion:** Though, GSLV-MK-II has faced some challenges due to a reverse engineered cryogenic engine, now India has developed its own indigenous cryogenic engine CE-20 which has proved very successful in all the launches of LMV-3 so far.

3) GSLV MK-II (LVM-3)

- Example Questions

- Discuss the significance of GSLV-MK-III in India's space program [10 marks, 150 words]

- LVM3 is a **3-stage** heavy lift launch vehicle developed by ISRO.

- **Capability:** GSLV-Mk III can launch 4 tons class of satellites to Geosynchronous Transfer orbit (GTO) or about 8-10 tons to LEO, which is twice the capability of GSLV Mk II.
- **Flights so far:** GSLV-MK-III D1 (GSAT-19); GSLV-MK-III D2 (GSAT-29); GSLV-MK-III-M1 (Chandrayaan-2); GSLV MK-III-M2 (OneWeb India-1); LVM3-M3 (OneWeb India-2); LVM3-M4 (Chandrayaan-3)

- **Significance of GSLV MK-III for India's Space Program**

- **Self-Reliance:** The vehicle has made ISRO fully self-reliant in launching heavier communication satellites of INSAT-4 class, which weighs between 4500 to 5000 kg.
- **Saves Money:** The mission's success will enable India to launch four-tonne satellites on its own rocket instead of paying huge amounts of money to foreign space agencies to execute the operation.
- **Commercialization:** Launching satellites is a huge market, and India's PSLV has already shown its mettle. Now it's time for GSLV to show its significance.
- **Advancement in Science and Technology:** The development of GSLV-MK-III enhances India's image as a Space Power. The latest cryogenic engine CE-20 is another of India's S&T achievements.
- **Enhances India's Soft Power:** We will be able to launch heavier satellites for other countries and this will play a big role in improving our significance at the world stage.
- **Future ISRO Missions:** ISRO in long term is also planning GSLV-MK-III to be India's first vehicle to ferry people into space (Gaganyaan).

- Conclusion

- Successful launch of OneWeb India-1, OneWeb India-2, Chandrayaan-2 and Chandrayaan-3 on board LVM-3 is a feather in ISRO's cap. The rocket has the potential to change the satellite launch landscape for India and other countries who take space services from India.

4) SMALL SATELLITE LAUNCH VEHICLE (SSLV)

- **Practice Question:** Why does India need to develop an SSLV when ISRO already has a very successful PSLV and a very powerful GSLV [10 marks, 150 words]
- **Introduction**
 - The Indian SSLV (Small Satellite Launch Vehicle) is a small launch vehicle which will serve small satellite launches. It is the **smallest vehicle** at 110-ton mass at ISRO.
- **Key Features:**
 - **Capability:** Payload capacity of 500 Kg to 500 km planar orbit or 300 kg to SSPO. Using PSLV for these small satellites was an overkill.
 - It uses **three solid fuel-based stages** and a liquid fuel-based velocity trimming module (VTM) to place the satellite in orbit.
 - **Major technologies** developed as part of SSLV are flexible nozzle control with electro-mechanical actuators for all stages, miniaturized avionics, and a velocity trimming module in the upper stage for precise satellite injections.
- **Need:** The **global demand** for launch of small satellite is increasing. It is being demanded by businesses, government agencies, universities, and various research labs.
- **Advantages:**
 - **Low turnaround time:** It is ideal for on-demand, quick turn-around launch of small satellites. It will take only 72 hours to assemble (unlike around 70 days needed for PSLV).
 - **Low Human Resource requirement:** Only 6 people will be required to do assembly (unlike 60 people for the PSLV).
 - **Cost Effective:** The overall cost of building the SSLV will only be Rs 30 crores.
 - **More business opportunities.**
- **Conclusion:** The successful completion of 2nd developmental flight (SSLV-D2 EOS-07) has set the foundation for the future launches of SSLV. It is going to be an important launch vehicle in the ISRO's pool for commercial launches.

5) FUTURE OF INDIA'S LAUNCH VEHICLES

- **Introduction:** ISRO currently has four categories of launch vehicles. This include SSLV, PSLV, GSLV-MK-II and LVM3. But despite this, India is dependent on other countries for launch of communication satellites of 5 ton plus weight.
- **To increase the capability of ISRO launchers**, following are the key initiatives ISRO has planned:
 - » **Next Generation Launch Vehicle (NGLV)** is being developed in which ISRO is understood to be looking at a cost efficient, three stage rocket, which will be a reusable vehicle with a capability of sending 10 ton of payload to GTO.
 - » **NGLV** is planned to feature Semi-Cryogenic engine in the second stage which will enhance the capability of the overall rocket.
 - » It is also expected to have simple robust design which will allow bulk manufacturing and modularity in systems, sub-systems leading to very limited turnaround time.
 - » **ISRO Tested Hybrid Propulsion System:** It uses solid HTPB and Liquid Oxidizer (LOX) as fuel and thus gives the solid fuel stage the ability of maneuverability.
- **Conclusion:** While ISRO has already emerged as one of the top five space powers in the world, in the next decade, it is aiming to consolidate its position among the top 3 countries. The above initiatives will be positive steps in the direction.

6) SATELLITE NAVIGATION SYSTEM (SATNAV SYSTEMS)

– Some Practice Questions:

- What is satellite navigation? Discuss the key economic and social uses of Satellite navigation. [10 marks, 150 words]
- Why did India need its own regional satellite navigation system when we already had access to GPS services? Discuss the key features of NAVIC. [10 marks, 150 words]

- **Introduction:** A satellite navigation (SATNAV) system is a technology that allows users to determine their precise location, velocity, and time information anywhere on or near Earth's surface. It uses a network of satellite in space and provide accurate positioning data.

- **Methods used in SATNAV: Triangulation and Trilateration**

- **Applications: Navigation purposes** -> helping users find their way while driving, hiking, or boating. It is also used in aviation, surveying, geolocation-based services, precision agriculture, and even in some outdoor recreational activities. It can be used for vehicle tracking, fleet management, precise timing etc.

- **Global Positioning System (GPS)** is the best-known satnav system, GPS, uses 24 active satellites (including backups). Day and night, 365 days a year, they whiz around earth once every 12 hours on orbital plane inclined 55 degrees to the equator. It uses the principle of Trilateration to provide the services.

A) NAVIC (NAVIGATION USING INDIAN CONSTELLATION)

- Indian Regional Navigation Satellite System (IRNSS) (also called Navigation Using Indian Constellation (NAVIC)), is a regional satnav system developed by ISRO. It aims to provide reliable position, navigation and timing (PNT) services over India and its neighbourhood, upto 1500 km from its boundary. In addition it is also capable of broadcasting messages. This can be used for broadcasting safety-of-life alerts in areas with poor or no communication, particularly in Ocean.
- **Need of IRNSS** when services like GPS are easily available.
 - The access to foreign controlled global navigation satellite systems is **not guaranteed in hostile situations**, as happened to Indian military depending on American GPS during **Kargil War**.
- **NAVIC provides two types of services:**
 - **Standard Positioning Service** (Open for Civilian Use)
 - **Restricted Services** (Encrypted one, for authorized users (military))
- **Components of IRNSS System:**
 - Space segments consists of **7 satellites, 3 satellites in GEO stationary orbit (GEO)** and **4 satellites in GEO synchronous orbit(GSO)** with inclination of **29 degree** to the equatorial plane.
 - All the satellites will always be visible in the Indian region.

FIRST OF THE 2ND GENERATION SATELLITE – NVS-01 WAS SUCCESSFULLY LAUNCHED IN MAY 2023

- ISRO's **GSLV F12** (GSLV-MK-II mission) successfully places navigation satellite NVS-01 into intended orbit.
- **About NVS-01:**
 - » **Heavier:** It weighs 2232 kg and has been placed in geosynchronous orbit (older IRNSS satellites weighed 1,425 kg)
 - » **Indigenous Atomic Clock:** For the first time, the satellite carries an indigenous atomic clock. The space qualified Rubidium atomic clock has been indigenously developed by Space Application Centre – Ahmedabad.
 - » **L1 signals for better use in wearable devices:** The second generation satellites have send signals in a third frequency, L1, besides the L5 and S frequency signals that the existing satellites provide. This will increase operability with other satellite based navigation systems. L1 frequency is the most commonly used in the GPS and will increase the use of NavIC in wearable devices which use low power signal frequency chip.
 - » **Longer Mission Life** of 12 years (earlier NavIC satellites have a mission life of 10 years).

Where is NavIC being used?

- **Initial Delay in developing user receiver:** A 2018 report by the CAG of India has said that even though the cabinet cleared funding of Rs 200 crore to develop user receivers in 2006, work on the project started only in March 2017, by which time seven launches of NavIC was already done.
- **But, now,** the receivers have been deployed, and NavIC is in use for projects like public safety, power grid synchronization, real-time train information system, and fishermen's safety. Other upcoming initiatives (such as) common alert protocol based emergency warning, time dissemination, geodetic network, unmanned aerial vehicles are in the process of adopting NavIC system.
- Some cell phone chipsets build by Qualcomm, MediaTek integrated NavIC receivers in 2019. Some example phones which are NavIC enabled include Redmi Note 9, Realme 6, the OnePlus Nord etc.
- **Way Forward:**
 - **Promoting the Use of IRNSS:**
 - **NavIC chip** -> affordable
 - **Spreading awareness** -> Positional accuracy better than 20 m and timing accuracy better than 50 ns (20);
 - **Make compulsory** for phones marketing in India: Mobile phones haven't been made compatible to process its signals.
 - **Government apps** -> start using NavIC for various purpose.
 - **HEIs** -> Promote use in Labs, among students etc.
 - **Fast track** the plans for making NavIC global
 - The next generation NVS series of satellites will make the NavIC more versatile and promote ease of use.

7) CHANDRAYAAN 3.0 (LVM3-M4 MISSION)

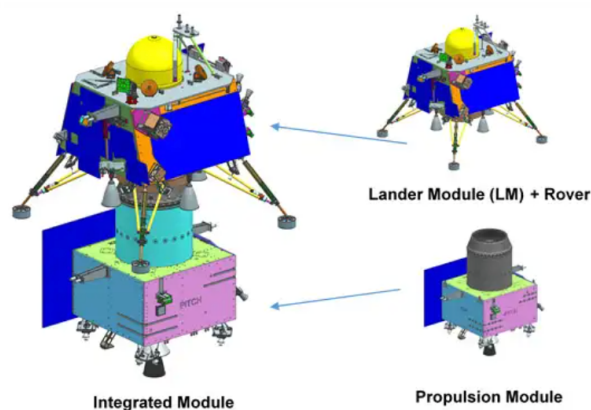
- Introduction: Chandrayaan -3 is the third Moon Mission of ISRO that was launched in July 2023 perched on GSLV-MK-3 heavy lift vehicle. It is a follow-on mission to Chandrayaan-2 and demonstrated end-to-end capability in safe landing and roving in lunar surface when it landed on the south pole of Moon on 23rd Aug 2023. With this, India has become the fourth country in the world after USA, Russia and China to successfully land on Moon.
- **Components of Chandrayaan 3.0:**
 - It consisted of a **Propulsion Module (PM)**, **Lander Module (LM)**, and a **Rover** with an objective of developing and demonstrating new technologies required for inter-planetary mission. **Note:** It doesn't have an orbiter module.

Propulsion Module (PM)

The main function of PM is to carry the LM from launch vehicle injection till final lunar 100 km circular orbit and separate LM from PM.

This propulsion module has Spectro-Polarimetry of Habitable Planet Earth (SHAPE) payload to study the spectral and Polarimetric measurements of Earth from the lunar orbit.

Note: Lander module separated from propulsion module on Aug 17, 2023. Here propulsion module was still left with 150 kg fuel. So it did something interesting, it was able to return to orbit around earth thus doing something which will be needed in future Chandrayaan mission.

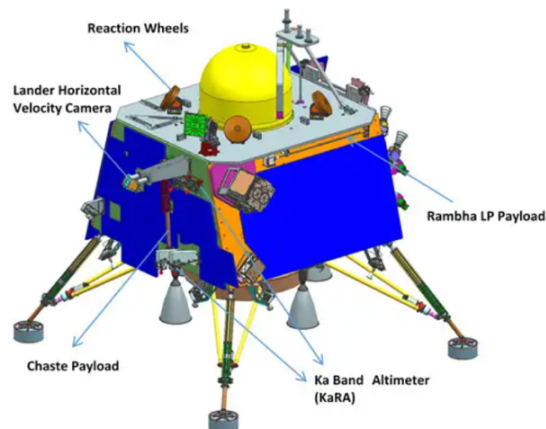


The Lander (Vikram) had the capability to soft land at a specified lunar site and deploy rover. It happened on 23rd Aug 2023. It remained stationary on the surface and carries four payloads which would record the chemical, thermal, and seismic instruments of the moon's surface.

Lander Payloads: Lander module has four payloads (Chaste, RAMBHA, ILSA and LRA)

Chandra Surface Thermophysical Experiment (ChaSTE): To carry out the measurements of thermal properties of lunar surface near polar region.

Instrument for Lunar Seismic Activity (ILSA) for measuring the seismicity around the landing site and delineating the structure of the lunar crust and mantle.



A passive Laser Retroreflector Array (LRA) from NASA is accommodated for lunar laser ranging

RAMBHA- LP (Radio Anatomy of Moon Bound Hypersensitive ionosphere and atmosphere) - RAMBHA:

studies. It acts as a target for lasers for very accurate measurements for future missions.

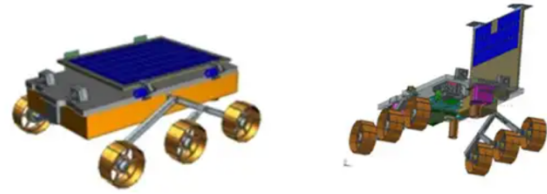
Rover (Pragyaan) is a 6 wheeled robotic vehicle.

Life: One lunar day (14 earth day)

Payload:

Laser Induced breakdown Spectrometer (LIBS): It will determine the chemical and mineral composition of the lunar surface.

Alpha Particle X-Ray Spectrometer (APXS): It will determine the composition of elements such as magnesium, aluminium, silicon, potassium, calcium, titanium and iron in the lunar soil and rocks.



▫ **Mission objectives of Chandrayaan 3.0**

- To demonstrate safe and soft landing on Lunar Surface
- TO demonstrate Rover roving on the moon
- To conduct in-situ experiments.

A) LANDING WAS THE MOST COMPLICATED PART HERE:

- Landing is the **most complicated part** of the mission. The Lander and Rover get ejected at a speed of around 6,000 km/hr and have to be slowed down to roughly 3 km/hr before it lands. Since moon doesn't have atmosphere, parachute kind of mechanism can't be used. Here, **thrusters** had to be fired in opposite direction to slow down the lander.

B) WHERE DID LANDER LAND?

- i. It landed at around 70-degree S near the southern pole of the moon.
- ii. **Why?**
 - a. The site was selected as there are several craters here that are permanent in shade and can be reservoir of frozen water which is key to the future space mission.

COMPARING CHANDRAYAAN-1, CHANDRAYAAN-2 AND CHANDRAYAAN-3

	Chandrayaan-1	Chandrayaan-2	Chandrayaan-3
Year	2008	2019	2023
Rocket Used	PSLV	LVM-3	LVM-3
Payloads	Orbiter + Impactor Module (for crash landing)	Orbiter + Lander (Vikram) + Rover (Pragyan)	Lander + Rover
Successful	Yes	Partially Yes (Lander failed)	Yes

	Perhaps the <u>most important discovery of Chandrayaan-1</u> was the <u>discovery of water and hydroxyl (OH) molecules</u> in the Moon's thin atmosphere (exosphere) as well on the lunar surface. <u>Buried Lava Tubes:</u> The terrain mapping camera and hyperspectral imager on board Chandrayaan-1 detected an <u>underground lava tube</u>, which, scientists believe, can provide a <u>safe environment for human habitation in the future</u>. It can protect against hazardous radiation, small meteoric impacts, extreme temperature and dust storms on the surface.	It helped in <u>separately identifying the water and hydroxyl molecules, and mapping water features across the moon for the first time</u>.	
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D) SIGNIFICANCE OF GOING TO MOON:

- It underlined India's rise as a space and technology powerhouse. It will also strengthen India's soft power in the global community.
- Since moon is the closest cosmic body to Earth, the plans to explore rest of the universe starts with exploration of the moon. Moon can also act as a **promising test bed to demonstrate technologies required for future deep-space missions**.
- It would further help **"stimulate the advancement of technology, promote global alliances and inspire a future generation of explorers and scientists."**
- **Resources:** Recent increase in interest in Moon is primarily due to possibility of important minerals being found on Moon.

WAY FORWARD:

- While the current achievement of being the first country to land on the south pole of the moon is commendation, India still needs to do a lot to compete with bigger space giants like USA, and China.
 - 1) **Look Beyond frugal engineering** as it is no longer enough to make a difference to global activity on the moon. To do more impactful moon projects, ISRO needs bigger budgets and more powerful rockets that can arrive quicker and with heavier payloads too.
 - 2) **Encourage bigger contribution from private sectors:** Globally, space programs are being animated by commerce, and private sector is emerging as bigger player than state (E.g SpaceX)

- 3) **Increased International Cooperation:** Joint programs can be economical, have shorter gestation period, and higher science outputs. Even NASA (USA) has realized that massive space program would not be feasible by external support and hence have launched initiatives like Artemis Accord. India also needs to raise its lunar profile through international cooperation.
 - 4) **Keep Geopolitics in Mind:** Both USA and China have drastically increased their interest and investment in Moon. USA plans to send humans again to moon in 2025 and China plans to do the same before 2030. Given India's tensed relations with China, India needs to embark on serious negotiations with the USA on the possible terms of mutually beneficial Indian participation in the Artemis Mission.
 - 5) **Getting domestic and international legal framework ready:**
 - India needs to follow up its space policy with legislation that facilitate and regulates space business.
 - India also needs to pay serious attention to shaping the global governance of space. The existing international regimes like Outer Space Treaty, 1967 and the 1979 Moon treaty are under considerable stress. Therefore, India should join hands with like-minded nations to reform the current outerspace order.
- **Conclusion:** Chandrayaan-3's success will inspire generations of scientists and engineers – and all those who seek knowledge – to set the bar higher. For India, the moon is not the destination. It is a **springboard**.

8) INDIA'S OWN SPACE STATION PLAN

- **Example Questions**
 - What is India's plan to have its own space station and how will it benefit our space program? [10 marks, 150 words] [Mains 2019]
 - Discuss the key advances that India will need to achieve before developing the ability to have its own space station. [12.5 marks, 200 words]
- **Introduction**
 - A space station is a spacecraft capable of supporting crew members, designed to remain in space for an extended period of time and for other spacecrafts to dock.
 - Currently, there are only two fully functional space station in the earth's lower orbit.
 - **The International Space Station:** It is a partnership between NASA (USA), JAXA (Japan), CSA (Canada), and Roscosmos (Russia). It is world's largest international cooperative program in science and technology.
 - **The Tiangong Space Station:** China is also building a space station of its own.
 - **India** is also plans to do the same and with this in mind ISRO plans to have its own space station. **PM Modi** has directed the ISRO to aim for setting up "Bhartiya Antariksha Station" by 2035.
 - The proposed space station is envisaged to weigh **20 tonnes**, which will be placed in LEO at a height of 400 km and will serve as a facility where astronauts can stay for 15-20 days.

- India will become only the fourth country to launch a space station after US, Russia and China which plans to launch it in 2020.

▫ **Advantages/Benefits/ Significance**

- **Natural Progression** for any space faring country after Human Space flight.
 - The Human Space flight leads to development of human rated rocket, the ability to train crew and the capacity to sustain life and safety of the crew on orbit and return them.
- **Step towards** future manned expedition to other celestial bodies.
- Ability to do **micro-gravity experiments.**

▫ **What advancements will India need to achieve to have its space, Stations?**

- All the **Gaganyaan requirements** (Space suits, Training facilities for astronauts, Crew Escape Module, making GSLV Human rated, developing a habitable module etc)
- **Larger bigger rocket** by upgrading the capabilities of GSLV-MK-III (it is right now capable of carrying on 10 tonnes to LEO)
- Developing ability to perform **space docking**. ISRO has revealed its plan to carry out a space docking experiment, Spadex.
 - Two experimental modules will be sent into space, on board a PSLV rocket, and two will be docked with each other.
 - Complex docking technology is an essential building block process for setting up the country's own space station and sending humans to that station.
 - The docking process involves latching of one satellite with another with the help of on-board cameras, and constant monitoring by the earth's station. The complexity of the process comes as it involves controlling the speed of the two satellites and bringing them together so that they can dock and become a larger structure.
 - It can serve two goals: To send humans from a space shuttle to a space station and to assemble a larger satellite in space.



▫ **Conclusion**

- Though the Space Station program is still some distance away, by initiating the steps such as sending humans to space, conducting the SPADEX experiment etc ISRO has started making significant moves towards this very gargantuan and challenging task.

9) DR VIKRAM SARABHAI AND HIS CONTRIBUTIONS (12TH AUG 1919 – 30TH DEC 1971)

- Vikram Sarabhai, **father of Indian Space Program**, was born on 12th of Aug, 1919 in Ahmedabad.
- **Key contributions**
 - He was a great institution builder and established or helped to establish a large number of institutions in diverse fields. He established **Physical Research Laboratory (PRL)** in 1947. PRL was the cradle of space sciences in India. PRL had a modest beginning at his residence, the RETREAT, with research in cosmic rays. It was formally established at M.G. Science Institute, Ahmedabad, on 11th Nov 1947
 - He played an important role in establishment of a number of institutions including IIM Ahmedabad.
 - The **establishment of ISRO** was one of his greatest achievements. He successfully convinced the government of the importance of space program after the Russian Sputnik launch.
 - He was the first chair of Indian National Committee for Space Research (INCOSPAR) which was predecessor to ISRO (established in its current form in 1969).
 - He also contributed in the setting up of Thumba Equatorial Rocket Launching Station at Thiruvananthapuram, with its inaugural flight in 1963.
 - He was also chairperson of Atomic Energy Commission.
- **Recognition**
 - i) He received Shanti Swarup Bhatnagar Medal in 1962
 - ii) Was awarded Padma Vibhushan (posthumously) in 1972. Earlier was awarded Padma Bhushan in 1966.
- **Other Key Positions held:**
 - i) President of the Physics Section, Indian Science Congress (1962)
 - ii) President of General Conference of the I.A.E.A, Vienna (1970)



10) COMMERCIALIZATION AND PRIVATIZATION IN SPACE SECTOR

- In June 2020, government approved the participation of private players in all sectors of space activity – earlier they could only be vendors/suppliers.
- **NSIL** – Established in March 2019 to commercially utilize R&D work of ISRO Centres and other constituent unit of DOS.

- **In-Space (Indian National Space Promotion and Authorization Centre)**, set up in 2020, is an independent nodal agency under Department of Space (DoS) to boost commercialization of Indian Space Activities and encourage private players.
- **The Indian Space Policy 2023** was notified as an overarching, composite and dynamic framework to implement the vision of unlocking India's potential in space sector **through enhanced private participation**.
- **Amendment in FDI Policy in Space Sector:**
 - Earlier, FDI is permitted in establishment and operation of satellites through the government approval route only.
 - But, through the 2024 amendment in FDI policy in space sector, FDI policy on space sector has been eased. **Now 100% FDI is allowed in space sector**. The entry route for the various activities under the amended policy are as follows:
 - 1) **Upto 74% under Automatic route**: Satellites-Manufacturing & Operation, Satellite Data Products and Ground Segment & User Segment. Beyond 74% these activities are under government route
 - 2) **Upto 49% under Automatic Route**: Launch Vehicles and associated systems or subsystems, Creation of Spaceports for launching and receiving Spacecraft. Beyond 49% these activities are under government route.
 - 3) **Upto 100% under automatic route**: Manufacturing of components and systems/ sub-systems for satellites, ground segment and user segment.
- **Positive Impact:**
 - Skyroot Aerospace (based in Hyderabad) has already launched VIKRAM series of rockets.
 - **Agnikul Cosmos**, a Chennai-based start-up, is planning the commercial launch of its rocket Agnibaan in the first quarter of 2023.
- **Way Forward to further promote Private Sector:**
 - Identify and encourage more private players by enabling policy environment.
 - **Existing big players in space** like Godrej & Boyce, Larsen & Toubro, and Walchandnagar Industries are among many private sector companies that have contributed to Indian Space Program. Government should work towards activating their full mobilization into the India Space Effort. It is these large companies that will bring **much needed scale to make India a Economic Powerhouse**.
 - **International Cooperation and Investment** would be crucial for growth of private sector. Even NASA is working on Artemis mission in collaboration with a number of other space agencies.
- **Conclusion:**
 - **Skyroot Aerospace, Agnikul Cosmos etc** are just the beginning. More private players are needed in the space sector to meet the growing and diverse demand of India's Space Program.

11) REFORMING SPACE SECTOR IN INDIA

- **Need of Reforms:**

- **Low Global Space Economy Share:** Global space economy is currently valued at about USD 360 Billion. India, despite being among the few space-faring nations of the world, has only 2% of the global share.
 - **Negligible contribution of private sector:** In major space faring nations, over the last 2 decades several major private players have emerged who are bringing efficiency and innovation. For e.g. SpaceX, Virgin Galactic, and Blue Origin in USA. However, in India the role of space sector has been minimal, either being vendors or suppliers to the ISRO programs.
- **Guiding Principles of Reform:**
 - **Enables and Promote Private Players** – Level Playing Field, Favorable Regulatory Environment.
 - **Open Up ISRO Infrastructure and Technologies** – The reform also aims to make national space infrastructure developed over the years, available for use by private industry through business-friendly mechanism.
 - **Public sector to focus on R&D** – Let business entities focus on manufacturing and commercial activities. Previously developed and already mature technologies/platforms would be transferred to the private sector through Transfer of Technology mechanism.
 - **Demand Driven Approach for Development of Space Assets:** Optimize the utilization of space assets such as satellites and launch capacity by determining accountability amongst various stakeholders. Creation of new asset should be contingent on confirmation of demand from user agencies/ entities.
 - **Inspire Youngsters and Dreamers:** Develop world class learning facilities and space museums where youngsters can learn basics of space technology and carry out research in topics of interest.
- **Implementation Strategy:**
 - **Opening up of Space Sector**
 - **Setting up of IN-SPACE** (Indian National Space Promotion and Authorization Centre)
 - It is a single window, independent, nodal agency.
 - Its main mandate is to promote and enhance the role of private industry players in the space sector through hand holding, support, and by providing them with a level playing field.
 - **Provide a stable regulatory and policy environment:** The reforms have strengthened the policy making capacity of Department of Space and an exercise has been initiated to create new business friendly policy frameworks in areas like remote sensing, satellite communication, and launch policies.
 - **Enhancing the Role of New Space India Limited (NSIL):** The reforms have authorized the public sector NSIL to act as a public sector aggregator for both demand and supply of space assets.
 - **Transfer of Technologies from Public to Private Sector:** NSIL has been given the role of transferring technologies/platforms from the public to the private sector. Platforms such as PSLV and SSLV have been identified for transfer of technology to the private sector in the near future.

- **Announcements of future opportunities for private sector:** The reforms have tasked ISRO with identifying and announcing the future opportunities in selected science and exploration missions for private sector participation
- **Impact of Reforms:**
 - Private sector, industry, academia, and startups have warmly welcomed the space sector reforms and the new IN-SPACE mechanism.
 - More than 40 proposals from startups, MSMEs, and industries have already been received for future consideration by IN-SPACE, illustrating the overwhelming response to the system.
 - Increased Investment – Following these reforms, several Indian space sector startups have been able to raise venture capital for their planned project. A new Indian Space Association has been created to function as the advisory and advocacy group for the space industry.
 - 6 space technology incubation centres are now operational at Agartala, Trichy, Jalandhar, Rourkela, Nagpur, and Bhopal.
 - In Feb 2021, NSIL conducted its first commercial launch. The launch put 19 satellites into orbit on the PSLV C-51 launch vehicle, including 4 satellites through IN-SPACE.

12) REGULATION OF GLOBAL SPACE SECTOR

» Example Questions

- “Technological changes have made the global space sector full of future potentials, but limitations of multilateral governance framework hinder it” Discuss [15 marks, 150 words]

» Introduction:

- Outer space has become an arena of some of the most memorable technology demonstration.
- **Growth of Space Industry:**
 - Technological changes augur well for space industry. SpaceX's Falcon 9 sends satellite to space at the cost of \$2720 per kg. This is expected to go down to \$100 per kg in next decade. This will lead to increase in number of satellites and human tourists going to space.
 - According to Bank of America Report, today the space market is of \$350 billion, and this is expected to go to \$2.7 trillion by 2050.
 - Private sector's contribution in growth of space sector would be immense. SpaceX's Starlink plans more than 10,000 satellites to provide everyone everywhere access to internet.

» Key challenges in fulfilling Global Space Potential

- **Multilateral Governance Framework of outer space is obsolete:** Space law is the product of 1960s and 1970s.
 - **The Outer Space Treaty of 1967** enshrines the idea that space should be “province of all mankind” and “not subject to national appropriation by claims of sovereignty”. **The Rescue Agreement, Space Liability Convention, and the Space Registration Convention** further

expanded the provisions of the outer space treaty. The **Moon Treaty** of 1979 was not ratified by the major space faring nations.

- **Key Problems:**

- No Dispute Resolution Mechanism
- No provisions related to collision or debris.
- Insufficient guidance on interference with other's assets

- These gaps heighten the potential of conflict in an era of congested orbits and breakneck technological change.

▫ **Legal Frameworks are state-centric and don't cover private sector.**

- However, the non-state entities are now in the fray for commercial space exploration and utilization.

▫ **Space Arms Race is difficult to curb.**

- Militaries rely a lot on space technologies. So, investment in technologies that can destroy or disrupt space-based capabilities is under way.
- Not much has been done on this front. The UNGA passed a resolution on Prevention of an Arms Race in outer Space since 1982. But not much progress has been made since then. Further, the current geopolitical situation doesn't hold hope for addressing concerns of a space arms race.

» **Situation in India:**

- India has invested enormous resources in space sector through ISRO. Our future plans are also ambitious which include sending humans to space, establishing a solar observatory, installing a modular space station etc.
- India is also promoting the involvement of private players and for this an autonomous body IN-SPACe (Indian National Space Promotion and Authorization Centre) under Department of Space for permitting a regulating activity in private sector has been established.

» **Way Forward:** The Space environment that India faces requires us to grow beyond meeting technical milestones. We need a space legislation enabling coherence across technical, legal, commercial, diplomatic and defence goals. Our space vision also needs to address the global governance, regulatory and arms control issues. As Space opens, our space vision needs broadening too.

13) AVOIDING WEAPONIZATION OF SPACE

- **Example Questions:**

- » Discuss the key applications of Space technology in the Defence sector. Suggest some measures to curtail the weaponization of space [15 marks, 250 words]

- **Introduction:** Space has **always been a militarized domain.** In recent years, military role of space has expanded across a full spectrum of relevant activities, perhaps, most importantly in positioning, navigation and timing (PNT) services; satellite communication; Space based intelligence, surveillance

and reconnaissance as well as targeting. In simple words, the modern militaries would be drastically weakened in the absence of support from space technologies.

- It is therefore, no surprise that most major powers including China, Russia, USA and India have either crossed or are poised to cross the rubicon to dramatically expand counter-space capability and develop new generation of advanced **anti-Satellite Capabilities**. Further, China, Russia and USA also have **ground-based counterspace systems**, such as uplink and downlink jamming, or as in the case of China, laser dazzling. Moreover, there is potential of cyber-attack on satellites or ground stations controlling them.
- This clearly shows that most of the major actors are moving towards a future where they assume that space would be a war fighting domain. This is expected to give rise to **rapid Space Weaponization** and **sharpening of risk of conflict and instability**. This is especially true since there is a potential for a '**grey zone**' action in orbit prior to a conflict.
- **Outer Space Treaty 1968 and its shortcomings**
 - » It forms the foundation of international space law and is attempt towards preventing the weaponization of space. It says that weapon of mass destruction (e.g. Nuclear Weapons) can't be deployed in space.
 - » But, it doesn't cover the non-nuclear space weapons. Also, the definition of non-nuclear space weapons is becoming a major hurdle as capabilities like cyber-attack; jamming of signals to disable space capabilities etc. are not specific to space.
 - » Finally, it doesn't provide a mechanism to deal with violations and inconsistencies.
- **Recent UN Efforts:**
 - » The UN General Assembly Resolution 75/36 on responsible behaviour in space represent the latest effort towards constraining a rush to weaponization of space. Though the resolution is non-binding, it tries to establish norms of responsible behaviour to allow further diplomatic progress that could make it legally more difficult to pursue space weapons. It sets a foundation that may eventually see banning of ASATs in future. But, here also major concerns emerge as to what constitutes ASAT capabilities. This is because Ballistic Missile Defence (BMD) may be used as ASATs. This was demonstrated by USA in **Operational Blue Frost** in 2008.
- **Way Forward**
 - » **Updated and Enhanced Space laws:** violation should quickly deliver diplomatic, economic and political consequence for a state which is violating the norms.
 - » **Regulatory reforms in international bodies** such as COPUOUS
 - » **Enhanced Space Domain Awareness:** An elaborate internationally trusted space domain awareness network should be created to verify parties remain compliant to international law and there is reduced risk of grey zone actions in space.

- Enhanced **deterrence by denial through space resilience**. This resilience can be brought by augmentation of satellites disaggregation of critical space support functions across larger numbers of small satellites, and the ability for rapid reconstitution of space capabilities – together with an inherent defensive retaliatory capabilities.

14) SPACE TOURISM

- **Why in news?**
 - » ISRO is planning space tourism by 2030
- **E.g. Questions:**
 - » What is Space Tourism? Enumerate some recent developments in the Space Tourism Sector. [10 marks, 150 words]
- **Space Tourism**
 - » Space Tourism is the segment of space travel which **provides non-astronauts the ability to go to space for recreational, leisure or business purposes**. The idea is to make space more accessible for anyone who can afford it.
 - » In the past, NASA and Russian Space Agency used to take tourists for space travel. For e.g. **Dennis Tito** was the first commercial spaceflight passenger before which only astronauts used to go to space. He went to space on Russian Soyuz TMA Launch Vehicle in **April 2001**. After him, between 2001-2009, few other space tourists went to space, aboard a Russian Soyuz space to ISS, brokered by Space Adventures (an American Space Tourist company) in conjunction with Roscosmos.
- **Recent tourism space flights:**
 - » **Virgin Galactic** is a company which was established by British Entrepreneur Richard Branson in 2004.
 - In July 2021, Richard Branson and five others undertook a brief trip to the edge of the space, taking off on a **VSS unity spaceship**.
 - » **Blue Origin** was established by Jeff Bezos in 2000. It's reusable rocket **New Shepherd** successfully completed first human flight to space recently (**20th July 2021**) with **four private citizens onboard** in Jeff Bezos. The flight went about **107 km** high.
 - » **SpaceX's Inspiration4** – debut of SpaceX's tourism business (Sep 2021)
 - Falcon 9 rocket took a crew Dragon spacecraft with 4 civilians (first all civilian space flight) into space. They travelled to an altitude of 575 km,

	BLUE ORIGIN	VIRGIN GALACTIC
Flies above the Kármán line (internationally recognized boundary of space, 100 km)	Yes	No
Vehicle type	Rocket	High altitude airplane
Windows	Largest windows in space 42" x 28" (107 cm x 71 cm)	Airplane-sized windows
Escape system	Yes	No
Ozone layer impact*	Minimal Exhaust is water with minimal impact on environment	High Hybrid rocket engine with HTPB & nitrous oxide, 100x more harmful
Flight history	15 Safe flights	3 Flights above 80 km

even higher than HST and ISS. In the orbits around the planet, they would see 15 sunsets and sunrise every day.

- Other than these three, companies such as Virgin Atlantic, XCOR Aerospace, Armadillo Aerospace are working on providing space tourism services to people.
- **Concerns**
 - » **Climate change** may be aggravated by Space Tourism.
 - There are concerns that space vehicles in future may become some of the biggest contribution in pollution and Climate change. It's because rockets emit upto 100 times more carbon dioxide per passenger than airplane.
 - An MIT study has shown that soot released from rocket launches are far more effective at warming the atmosphere when compared to other sources. This is because they emit gaseous and solid chemicals directly into the upper atmosphere. Further, **black carbon soot**, are almost **500 times more efficient at retaining heat**.
 - The study also showed that space tourism may undermine progress made by the Montreal Protocol in reversing ozone depletion. This is because the pollutants from rocket fuel and heating caused by spacecraft returning to earth, along with the debris caused by flights are specially harmful to the ozone layer.
 - » **Available only for highly rich people.**
- **Way forward**
 - » **Urgent need of environmental regulation** to reduce the climatic damage space tourism industry may cause.
 - » **Promote Green Space Tourism:** Blue Origins New Shepherd didn't produce any pollutant and these technologies need to be promoted in coming future for space tourism.
 - » **Make Space Tourism more inclusive** by making it affordable: As, the space industry grows, better efficiencies and scales would be achieved, making space travel for inclusive.
- **Conclusion**
 - » The recent efforts by the private space industry to increase access to space may become early pioneers in maritime exploration and aviation and may change the humankind forever.

15) SPACE BASED INTERNET

- **Example Questions**
 - What is Space based internet? What are its advantages over traditional fiber optics cables? Discuss how the SpaceX plans to provide affordable and non-stop internet throughout the earth. [15 marks, 250 words]

- **Introduction**
 - Internet has emerged as one of the basic human needs of the 21st century. But, even today more than 3 billion people don't have a reliable internet connection. The concept of Space based internet is aimed at changing this scenario.
- **Space X Plan**
 - **The Starlink Network** of SpaceX eventually plans to install 42,000 satellites to ensure non-stop internet services throughout the earth at a cost-effective rate. These satellites will be connected with their neighboring satellites using lasers.
- **China's "Guowang" (GW)** constellation has also been announced which is meant to meet satellite-based internet services.
- **Other such projects:** Several other companies including Amazon, OneWeb and O3B have also planned large constellation of satellites in LEO and MEO – but these projects are very small compared to Starlink.
- **Comparing Geostationary vs LEO satellites** for providing internet services [Advantages of LEO – Low latency-> allows real time communication; Disadvantage -> need more satellites for coverage as they cover small part of earth]
- **Significance**
 - The space-based internet services can provide reliable and uninterrupted internet services universally to everyone on earth.
 - **Services during emergency:** For e.g. During Russia Ukraine war in 2022, the Starlink played an important role in strengthening the Ukrainian military might after the European SATCOM system was cyber attacked.
 - **Services such as autonomous cars** will be made possible and things like IOT can be integrated in virtually every household.
 - Universal Internet access will also ensure better e-governance and thus enhance the ease of living for common citizens.
- **Concerns:** The ambitious space-based internet services (especially Starlink of SpaceX and Guowang of China come along with a number of downsides:
 - **Increased Space debris**
 - The SpaceX initiative thus will also **increase the chances of collision of satellites**. In fact, on Sep 2, 2019, only, ESA had to perform, for the first time ever, a 'collision avoidance maneuver to protect one of its live satellites from colliding with a mega constellation.
 - **Difficulty in Space Observation -> Light Pollution**
 - **Increased crowding in LEO and signal interference** in space may emerge as another problem.
- **Way forward**

- Establishing an **international convention that regulates the presence of satellites in LEO** is another possible way in making LEO less congested.
- All stakeholders need to jointly commit to reducing and mitigating the dangers posed by debris.
- **India** needs to plan for satellite-based internet system of its own.

- Conclusion

- Starlink is a very ambitious project and can solve a number of internet accessibility problems. SpaceX has already said that it is looking into reducing the challenges/limitations to as less as possible and it should always give these challenges a very high priority.

16) SPACE DEBRIS

▫ Introduction

- The term “space debris” refers to **defunct human made objects which are moving in orbit around earth.** It includes big and small things like discarded boosters, retired satellites, leftover bits and pieces from spacecrafts, screwdrivers, tools, nuts, bolts, lost gloves, flecks of paints etc.



- There are more than 20,000 pieces of debris that are larger than 5-10 cms and can be tracked and catalogued. There are hundreds of millions that we cannot because of their small size. They are all dangerous as they are moving at very high speeds.

▫ How are Space Debris created?

- **Breakup of older spacecrafts:** For e.g., breakup of US' spacecraft called USA 109 in 2015, created 100 debris pieces and 50,000 shards larger than 1 mm.
- **Accidentally left-over objects**
- **Testing of Space Weapons**
 - For e.g., China's testing of A-SAT missile in 2007 created more than 34,000 debris.
- **Further breakup of space debris:** More debris increase the chance of collision – a cascade effect known as the **Kessler Syndrome**. The fear is that the space could eventually become inoperable.
- **Mega constellations** (e.g., Starlink satellite internet constellation) would launch thousands of satellites in coming years and would make space more vulnerable to collision and debris creation.

- **Key Concerns Raised by Space Debris**
 - **Endanger the prospects for Space Missions** (Civilian, Commercial or military)
 - They are so fast moving that a fleck of paint can cause a crack in ISS.
 - Already instance of near-collision and maneuvering to avoid collisions have been noted. In 2019, European Space Agency (ESA) moved its earth observation satellite to avoid it colliding with Starlink satellite.
 - **Sometimes crash land on earth** harming life and livelihood of people
 - Recently parts of Zenit rocket debris are reported to have ended up crash-landing in Peru.
- **Key Steps being taken**
 - **Inter-Agency Space Debris Coordination Committee (IADC)** is an international government forum, founded in 1993, for the global coordination on the issue of man-made and natural debris in space.
 - ISRO is a member of the coordination committee along with other major Space agencies like NASA, ROSCOSMOS, ESA, UKSA etc.
 - **ESA's Clean Space Initiative** is studying an active debris removal mission called e. deorbit, which would target an ESA owned derelict satellite in low orbit, capture it, then safely burn it up in the controlled atmospheric re-entry.
 - **NASA's NanoRacks-Remove Debris:** It demonstrates an approach to reducing risks presented by space debris or "space junk".
 - **Japan Based Astroscale:** It is working to prevent space-debris collisions. It will use **magnet** to 'dock' orbiting junk circling the earth.
- **Some Concerns raised against projects for cleaning space Debris**
 - Some scholars feel that militarization of space technologies meant to clean up space debris poses hidden challenges.
- **Way Forward**
 - **We need to stop polluting:** We need to ensure that the new satellites are not leading to the problem.
 - **Remove the garbage.**
 - **Develop satellites** which can withstand the impact with small debris.
 - **Multilateral Regulations of mega constellation:** - Set up regulations and policies for mega constellations to ensure sustainable operations and to minimize risk of collision. This should also include new approaches for space traffic management as the capacity for ground-based platforms may not be sufficient to prevent collisions in case of increased number of satellites in future.
 - **Create code of conduct** for both public and private sector, regarding the responsible behaviours in outer space.
 - **Technology to deal with risk assessment, automate collision avoidance:**
 - Work towards reducing the risk of cyber security by risk assessment and effective risk mitigation policies.

- Use **advanced tech** like **ML, AI** etc to automate collision avoidance. E.g., ESA is developing an automated collision avoidance system.
- Above recommendations should be coupled with **responsible innovation** which is focused on use of technology for the benefit of international peace and security and not for weaponization.
- **Conclusion:** Keeping outer space safe, secure and sustainable is a joint global responsibility. For peaceful solutions to last, the deterrence and diplomacy, as well as public awareness, will have to be proactively forged by the world's space powers, leaders and thinkers.

3. ROBOTICS

1) SYLLABUS

- Awareness in the fields of Robotics

2) PAST YEAR MAINS QUESTIONS (SINCE 2013)

Q. What are the areas of prohibitive labour that can be sustainably managed by robots? Discuss the initiatives that can propel research in premier research institutes for substantive and gainful innovation [12.5 marks, 200] [Mains 2015]

3) OTHER PRACTICE QUESTIONS

1. Robotics in India is driven by private sector unlike rest of the Asia where government has played the most significant role in its growth. Give reasons. Discuss the initiatives that can propel robotics research in premier research institutes for substantive and gainful innovation [15 marks, 250 words]
2. "Robots are technological marvels, they cut errors, improve production and thus enhance efficiencies. But they are also the ultimate job stealers." In this light discuss the economic and political consequence of automation. [10 marks, 150 words]

4) INTRODUCTION

A) DEFINITION: ROBOTICS

Robotics is an **inter-disciplinary branch of engineering** that deals with the **design, development, operation, and application of robots** as well as computer systems for their control, sensory feedback, and information processing.

Robotics technology deals with **automated machines** which can replace humans in dangerous environments or manufacturing processes, or resemble humans in appearance, behavior, and/or cognition.

B) LAW OF ROBOTICS

Isaac Asimov gave the three laws of robotics as:

- A robot must not harm a human being, or, through inaction, allow a human being to come to harm
- A robot must always obey the human beings unless it is in conflict with the first law
- A robot must protect itself from harm unless it is in conflict with the first and/or second law.

5) APPLICATIONS: ROBOTICS IS BEING USED ACROSS A RANGE OF SECTORS SUCH AS:

- **Warehouse Automation**
- **Automotive manufacturing**
 - They add precision, tirelessness, and continuity in the manufacturing process.
- **Search and Rescue** after Disaster – Flood, building disaster, fires etc.
 - IIT Hyderabad is working on a search-and-rescue robot called **SARP (Snake like articulated platform)**. The engineering institute is applying several technologies in building **SARP**: navigation, camera, infrared, haptic feedback (to identify survivors) and collaboration (multiple snake robots can communicate with one another)
- **Defence & National Security**
 - Whether it is with Pakistan or China, our borders are unsafe for border forces and human lives are getting lost whenever there is firing or illegal movement of people at LoCs.
 - Government is looking at DRDO to develop next generation of robotic soldiers.
 - **Other dangerous security tasks** like **bomb disposal**, **reconnaissance** etc. can be performed by Robots.
 - For e.g., **Daksha** is one of India's current military robots. It is used to locate, handle, destroy, risky objects safely and even can climb stairs.
- **Hazardous Industries**
 - BARC is using robots to clean radioactive water tanks.
- **Mining and Mineral Extraction**
- **Marine Engineering**
 - **Amogh** is an autonomous underwater vehicle. It is designated to inspect and repair bridges, pipelines, and hulls of ships at the depth of upto 15 meters. The robot has an endurance capacity of upto 3 hours.
- **Space**
 - Robots are used for exploration when humans can't be used.

- E.g., Fedor of Russia, Vyomamitra of India etc.
- **Health Sector**
 - Robots are being used in operation theatres and rehabilitation centres to augment the quality of life.
 - Robots can also **assist surgical procedures** like removing gallbladders, performing hysterectomies, and repairing knee ligaments.
- **Agri-Robots**
 - Can replace costly human labor and add precision in agriculture.

They can be used anywhere to improve productivity

6) ADVANTAGES

- Accuracy
- Untiring nature of robots
- Non-complaining
- Safety in hostile environments
- Reducing cost of production
 - » Industries facing global competition can't survive without robots
 - » For e.g., the automobile industry can't actually survive without use of robotics

7) ROBOTICS IN INDIA

- Robotics technologies are all set to change the way things are done in the industries. It is also entering into the industrial space in India in a very fast manner.
- In India, robotics is being **driven by private sector** (unlike the rest of Asia where government has played a very significant role in promotion of robotics)
- In short, it is proving to be the **start-ups, local makers and foreign robotic companies that are creating the momentum in India**. They are cultivating the market and tapping opportunities.
- **Institutional Framework and Government Efforts**
 - **National Research Foundation**, an autonomous body under the new National Education Policy (NEP) 2020, has been established to boost research across segments, including AI.
 - **Centre for Artificial Intelligence and Robotics (CAIR)**
 - It is a laboratory of the DRDO, located in Bangalore, Karnataka.
 - It is mainly involved in the research and development of high-quality secure communication, command and control, and Intelligent System.
 - **'Make in India'** campaign provides a crucial opportunity for the robotics sector to flourish in India.

- **A Mission on Cyber Physical Systems** to support establishment of Centres of Excellence.
 - To invest in research, training and skilling in robotics, artificial intelligence, digital manufacturing, big data analysis, quantum communication and internet of things.
 - By DST.

A) CHALLENGES WHICH ROBOTICS SECTOR FACE IN INDIA

- i. **High cost of adoption - Capital Intensive nature of the sector**
- ii. **Scarcity of talent in the field - Human Resource**
- iii. **Lack of innovation**
 - Most of the projects being developed in India already exist in public domain
- iv. **Faculty/ research guides**
 - Scarcity of good quality faculty to guide research in this area
- v. **Industry:**
 - Absence of hardware companies that can cater to the industry and the dependence on countries like China, USA, and Europe to procure the necessary components is a major stumbling block.
- vi. **Lack of high paying jobs in India**
 - Lack of jobs in the field of robotics
 - Considering India's socio-economic structure - availability of jobs in the domain becomes an important factor for people working on a particular subject.
 - Not many companies have high quality jobs in India in the field of robotics and this leads to lack of enthusiasm towards the sector.
- vii. **SME find it difficult to see ROI before they proactively adopt Robotics**
 - This is also due to lack of examples or early adopters.
- viii. **Lack of support from government**
 - Rest of the Asian countries like China, Japan, Singapore etc., look to its government to define and dictate the development and direction of industry. Indian government has completely ignored the sector.

B) STEPS THAT NEEDS TO BE TAKEN TO PROMOTE ROBOTICS IN THE COUNTRY

- **Human Resource Development**
 - Promoting robotics from the school level so that more students choose robotics as career option
 - Improving the facilities at college level
 - Promoting robotics industry -> India's socio-economic condition forces students to participate in those industries where job opportunities are good.
 - Promoting Academy Industry collaboration

- Promoting International collaboration with countries which are advanced in robotics like South Korea, Japan and USA.
- **Industry**
 - Providing tax holidays for the Robotics sector
 - Improving ease of doing business
 - Ensuring encouraging environments for the entrepreneurs and startups.
 - Providing opportunities to private sector in the field of defence, space etc.
 - Lack of hardware components available in the country should be seen as an opportunity for entrepreneurs to set up businesses that can supply "Made in India" components and hardware to the existing and upcoming companies involved in Robotics.
- **International Collaboration**
 - More international collaboration and student exchange program in colleges
- **Attracting investment from Robotics industry in India:** Under make in India program

8) LIMITATIONS OF USING ROBOTS

- **Dislocation of Economy and job loss**
 - Former American president Barack Obama in his farewell Speech on 10th of Jan 2016 had said "next wave of economic dislocations won't come from overseas. It will come from automation that makes a lot of good middle-class jobs obsolete."
 - Robots are diminishing roles for unskilled labourers. India's largely uneducated labor force and broken education system aren't ready for the more complex jobs that workers need when their low-skilled roles are taken over by machines.
 - Meanwhile, nations employing robotics such as China, are becoming more competitive.
- **Increasing Inequality**
 - While jobs at the lower levels are at risk, most gains from automation will be concentrated to the rich rather than the poor.
- **Increasingly deadly wars**
 - Most of the scientifically advanced countries will be using robots and other automated weapons in future. This would make wars and conflicts more deadly and one sided.
- **Robots can become a danger for humankind**
 - In advanced forms robots using AI and ML may take over the decision-making power and thus can challenge the existence of humans.

9) ROBOTIC SURGERY

- Robotic surgery, or robot assisted surgery, allows doctors to perform many types of complex procedures with more precision, flexibility and control than is possible with conventional techniques.

- It is usually associated with minimally invasive surgery - procedures performed through tiny incisions. It is also sometimes used in certain traditional open surgical procedures.
- The surgeon uses controls in the console to manipulate special surgical instruments that are smaller, as well as more flexible and maneuverable than the human hand. The robot replicates the surgeon's hand movements, while minimizing hand tremors.
- **Da Vinci Surgical System**
 - It is a robotic surgical system made by the American company **Intuitive Surgical**. It is designed to facilitate complex surgery using a minimally invasive approach and is controlled by surgeon from a console.
 - The system is commonly used for prostatectomies, and increasingly for cardiac valve repair and gynecological surgical procedures.
- **Advantages**
 - It enhances precision, flexibility and control during operation and **allow doctors to better see the site**, compared with traditional system.
 - Robotic instrument has 360-degree rotation compared to the human hand which has only 270-degree rotation.
 - 10 times magnified, high definition, 3D-image of the body's intricate anatomy using key-hole techniques.
 - **Minimally invasive surgical operations** further have following benefits
 - Fewer complications such as surgical site invasions
 - Less pain and blood loss
 - Quicker recovery
 - Smaller, less noticeable scars.

4. NANOTECHNOLOGY

1) PAST YEAR MAINS QUESTIONS (SINCE 2013)

1. What do you understand by nanotechnology and how is it helping in health sector? (2020)
2. Why is nanotechnology one of the key technologies of the 21st century? Describe the salient features of Indian Government's Mission on Nanoscience and Technology and the scope of its application in the development process of the country (2016)

2) OTHER PRACTICE QUESTIONS

- a) What is Nanotechnology? Why is it called a general-purpose technology? [10 marks, 150 words]
- b) "Their unique nano-features may also make them potentially hazardous for human health and environmental safety" Discuss the key concerns associated with nanotechnology. [10 marks, 150 words]

- c) What are molecular machines? Discuss the significant role that it may play in various socio-economic sectors. [10 marks, 150 words]
- d) Discuss the key provisions of the Mission on Nanoscience and Technology [10 marks, 150 words]
- e) Write a short note on key properties and applications of:
- Graphene
 - Carbon Nano Tube (CNT)
 - Nanodrums
 - Schwarzites
- f) Suggest some measures to promote nano-technology research in the country. [10 marks, 150 words]
- g) “Nano Machines are going to have serious economic, social, environmental and military implications” Elaborate. How far does Indian Government’s Mission on Nanoscience and Technology prepare India for these implications? [15 marks, 250 words]
- h) What is Nano-UREA? How is it going to solve the various problems faced by fertilizer sector in India? [10 marks, 150 words]

- **Definition:**

- a) Nanotechnology is science, engineering, technology, conducted at the **nanoscale which is 1 to 100 nanometers**. Nanotechnology and Nanoscience involve the ability to see and to control individual atoms and molecules. In other words, nanotechnology is the engineering of the functional system at molecular scale.
- b) **Richard Feynman**, the father of nanotechnology, in his 1959 talk described nanotechnology as a field which can manipulate and control things on the scale of a nanometer. He expected that matter will have surprising properties at Nano level and thus provide for enormous number of applications

- **General Purpose Technology:**

3) APPLICATIONS

- **Carbon Nanotubes** are used in various products ranging from paints and textiles to medical diagnostics tools and components of future quantum computers because of remarkable properties such as very high elastic strength alongside low mass density or very high current densities with no heat loss.
- **Health**
 - **Nanoparticles of silver** embedded into fibers have anti-microbial action. It is used in food packaging, clothing, disinfectants, and household appliances. Bandages are being infused with silver nanoparticles to heal cuts faster.
 - **Gold Nanoparticles** have anti-bacterial properties
 - **Nanomedicines** – Diagnosis and Treatment (see details separately)
 - **Water Purification: Special Filters using nanomaterials** can remove objects as tiny as viruses from water.
- **Nanotechnology in Agriculture** – Better fertilizers, pesticides, insecticides, feeds, better treatment for domesticated animals.
 - **Nano-Fibre based Agriculture Inputs:** For e.g., **FIB-SOL provides** a five-gram fiber that is soluble in water and can be applied on field using conventional or modern irrigation practices. The product addresses the demand for live bacteria that could rejuvenate the soil. It could also increase the nutrient utilization efficiency, allowing plants to assimilate nutrients in a better way.
 - **Nano Urea and Nano-DAP** developed by IFFCO
- **Environmental Applications:**
 - **Iron nanoparticles** can be used to effectively clean-up organic solvents that are polluting the ground water. The nanoparticles disperse throughout the water and decompose the organic solvents in place.
 - Adding a **little boron to Carbon** while creating nanotubes produce solid, spongy, reusable blocks that can absorb large quantities of oil spilled in water.
 - **Nanotechnology based smart windows** have energy saving, easy cleaning, UV controlling and photovoltaic properties.
 - **Renewable Energy Generation**
 - New and Cheap Solar Cells use nanoparticles of **Titanium oxide** coated with dye molecules to capture the energy of visible light and convert it into electricity.

- A **novel catalytic nanosheet** from of a nickel molybdenum-nitride, a thousand time cheaper than traditional platinum, is the new model for harvesting hydrogen from water for use as fuel.
- **Electronics**
 - **Graphene** is used in transparent electrodes for solar cells, LCD, robust non-volatile atomic switches, chemical and biological sensors and in spintronic devices.
 - **Semiconducting nanowires** are highly versatile optoelectronic components, for a wide variety of applications such as nano-LEDs and nano-Lasers, solar cells, and biomedical sensors.
- **Structural Engineering Applications**
 - **Nano-enhanced Cement** contained by addition of nanoparticles like nano silica (silica fume), nanostructured metals, CNTs and carbon nanofibers give stronger, more durable, self-healing, air purifying, fire resistant, easy to clean quick compacting structure.
 - **Nano-enhanced Construction Ceramics** such as floor and wall tiles and sanitary ware have self-cleaning, anti-bacterial, hygienic, and scratch resistant features.
 - **Nano-enhanced paints** can reduce emission of Nitrogen-di-oxide, hydrocarbons, and carbon monoxide in the atmosphere. It can also make paint scratch proof, easy cleaning, air purifying, UV resistant, water repellent, flame resistant, and anti-bacterial.
 - **Nanotechnology based smart windows** have energy-saving, easy cleaning, UV controlling and photovoltaic properties.
 - **Fire resistant glasses** are produced by addition of fumed silica nanoparticles in glasses.
- Nanotechnology can promote **different forms of insulation solutions** like coatings, vacuum insulations glazing and nanofoams.
- **Nano whiskers** on clothes create a cushion of air around the fabric so that liquids can't stain them.
- **Cosmetics:** Nanoparticles like Zinc oxides and titanium oxides are used in sunscreen and related products. They provide protection from UVA rays.

4) FUTURE POTENTIAL APPLICATIONS

- Advancements like **nano-machines** will lead to development in the field of nano-medicines, more advanced electronics circuitry, quantum computing etc.

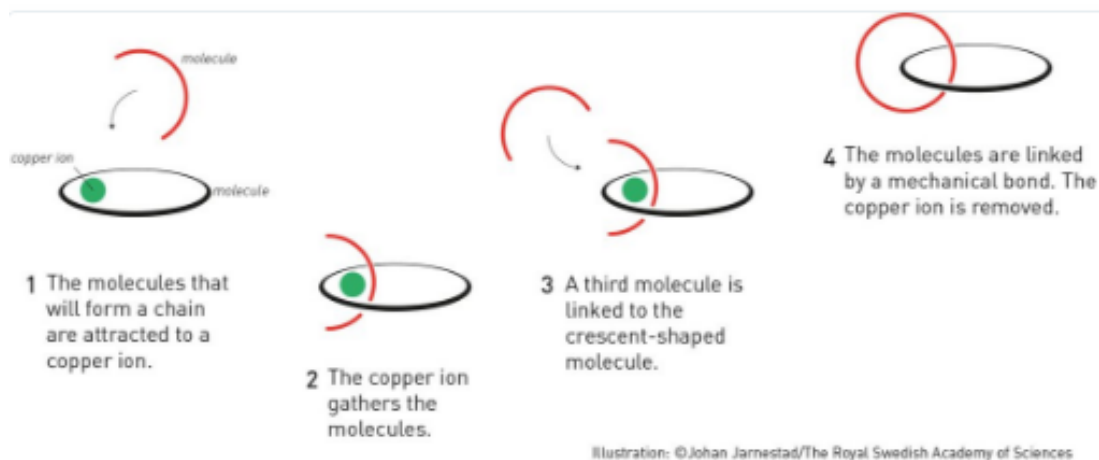
5) CONCERNS AND LIMITATIONS

- Due to their extremely small dimensions, large surface area and high reactivity, they have the **potential ability to penetrate living cells** quite readily. As a result, their unique nano-features may also make them potentially hazardous for human health and environmental safety.
- **Health**

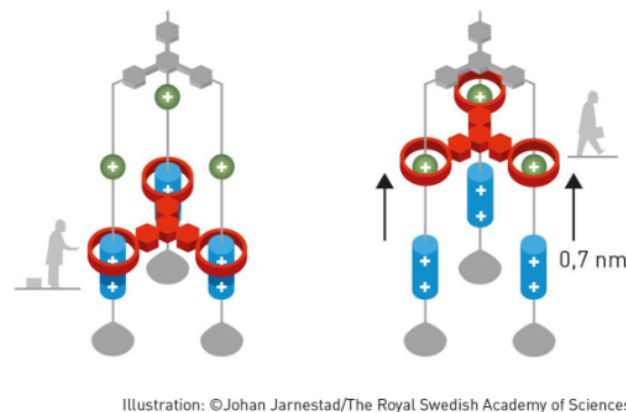
- Inhaling airborne nanoparticles and nanofibers may lead to a number of **pulmonary diseases**, e.g., fibrosis. Some form of carbon nanotubes could be as harmful as asbestos if inhaled in sufficient quantities.
- Experiments with rats have also shown impact on **skin (ageing) and brain**.
- **Toxicity**
 - Lack of investment on nanotoxicology research
- **Environmental impact**
 - **Lack of research on potential harmful impact:** Lack of study on Impact of nanomaterial on non-human species, on ecosystem or the global environment.
 - e.g. bacteriostatic **silver nanoparticles** used in socks to reduce foot odor are being released in the wash. These particles are then flushed into the wastewater stream and may **destroy bacteria which are critical component of natural ecosystem**, farms, and wastewater treatment processes.
- **More dangerous Weapons**
 - As a general-purpose technology, it will be **dual use**, meaning it will have many commercial uses and it also will have military uses - making for **more powerful weapons and tools of surveillance**.
 - A technology this powerful could easily be misused. The rapid development cycle and massive manufacturing capability may lead to an **unstable arms race** between competing powers.
- **Other Concerns**
 - May lead to **loss of jobs** in traditional farming and manufacturing sector
 - May bring about **crash in certain markets** due to lowering of oil and diamonds due to possibility of developing alternative source of energy that are more efficient and won't require use of fossil fuels. Also, because people would be able to develop products at molecular level, diamond will lose its significance.
 - Atomic weapons may become more accessible and more powerful and more destructive.

6) MOLECULAR MACHINES / NANO MACHINES

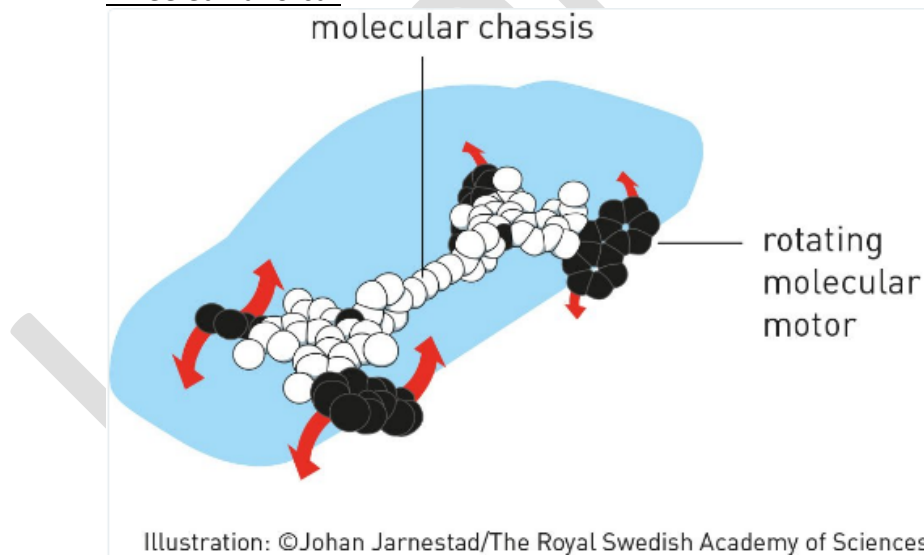
1. Molecular Machine, or nano-machine, is any discrete number of molecular components that produce quasi-mechanical movements (output) in response to specific stimuli.
2. **The 2016 Nobel Prize for Chemistry was awarded to 'Molecular Machine' trio for the design and synthesis of molecular machines**
 - **Details about their contributions**
 - a. **Sauvage** in 1983 took the first step by linking two ring shaped molecule to form a chain



- b. **Stoddart** in 1991 developed a **rotaxane**, a dumbbell-shaped molecular structure that enabled him to **build molecular lift**, a **molecular muscle** and a **molecule-based computer chip**



- c. **Feringa** in 1999 was the **first person to develop a molecular motor** and in **2011** designed a **four-wheeled nano-car**



3. Significance of those nano-machines

4. These **tiny machines** that we can't even see have **enormous potential**.

▫ **Medicine and treatment**

- Molecular technology could lead to development of machines that are **so small they could be swallowed or implanted into human bodies** with **little negative effect**.

- They could be used to fight disease in the body, to repair damaged tissues, and even to probe DNA structure.
- Such precise drug deliver will **minimize adverse side-effects.**
- **Smart materials** able to adapt to their environment, small sensors that can be controlled remotely, and drugs that are activated on command
- **Efficient energy storage devices**

7) EXAMPLE NANO-MATERIALS

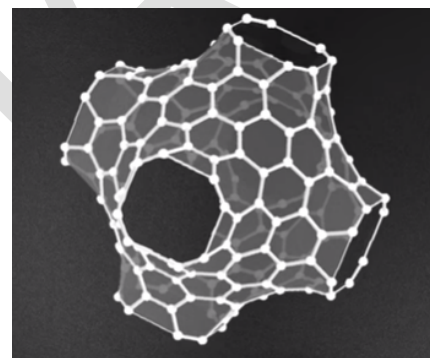
A) SCHWARZITE – NEW FORM OF CARBON CREATED

1. Example Questions

- What is a Schwarzites? Discuss its future potential applications [10 marks, 150 words]

2. Introduction

- **Schwarzites** is a completely new form of carbon that has been created by scientists recently. It is a **negatively curved** carbon unlike fullerenes and graphene which have positive or zero curvature respectively.



3. Creation of Schwarzites

- Scientists, in Japan and South Korea, while investigating Zeolite-templated carbons (ZTCs) – crystalline form of silicon di-oxide with carbon structure built into them – **accidentally created schwarzites.**
- Later, scientists at University of California demonstrated that it was possible to create ZTC structures by injecting a vapor containing carbon into zeolites.
- Once inside, the carbon assembles into a graphene-like sheet that lines the walls of the pores in the zeolite. In doing so, the surface stretches to minimize the area. As schwarzites are **topologically minimal surfaces**, they tend to acquire negative curvature. The zeolite is then dissolved to produce pure schwarzites.

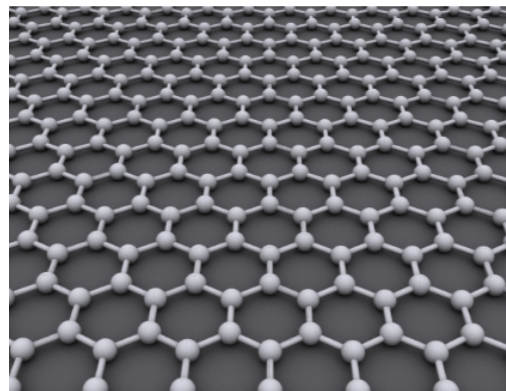
4. Potential Future Applications (details in class discussion)

- **Super Capacitors:**
- **Storage of atoms and molecules**
- **Catalyzing Reactions**

5. **Conclusion:** Though the applications of schwarzites remains to be seen, the scientists speculate that their unique electronic, magnetic, and optical properties can make them useful, as super capacitors, battery electrodes and catalysts, and with their large internal spaces ideal for gas storage and separation.

B) GRAPHENE

- It is an allotrope of carbon which is a one-atom thick layer of pure carbon. Carbon atoms are bounded together in a hexagonal honeycomb lattice.



- How is it produced**
 - By separating a single atom layer film from graphite.
 - Properties: Physical**
 - 2D** – world's first 2D material
 - Graphene is **harder** than diamond, **more elastic** than rubber, **tougher** than steel and yet **lighter than aluminum**.
 - In fact, it is 200 times stronger than steel (100 times stronger than the strongest steel).
 - Thickness:** 1 million times thinner than a human hair
 - Stretchable as well as transparent, flexible, and impermeable.
 - It can also act as **perfect barrier** – not even helium can pass through it
 - Properties: Thermal, Electrical and Magnetic Properties**
 - Highest electronic conductivity** of any material in the world.
 - Best Heat conductivity** of any material in the world
 - Shows a **large and nonlinear diamagnetism**
- Applications:** Graphene's unique combination of extraordinary properties offer a fascinating material platform for the development of next-generation technologies in many areas.
 - Energy Harvesting and Storage:** It can be used for better rechargeable batteries; superior capacitors; newer methods of making solar cells etc. Further, proton transfer in graphene shows promise for artificially mimicking photosynthesis.
 - Electronics:** Very high electron conductivity allows graphene to be used for low-cost printable electronics, high performance transistors; thermal management and heat dissipation in nano-electronic devices.
 - The optical properties** of graphene can also be controlled by doping and make it well suited for optoelectronic devices.
 - Composites and Coatings:** Its low mass and low loading requirements make graphene stand out as a reinforcing agent in composites. It can be used for making lubricants with enhanced anti-wearing capabilities; radiation shielding and lightning strike protection; superhydrophobic coating; transparent, flexible and conductive thin films etc.
 - Membranes** – It can improve the quality of filters used in desalination or other water purifying instruments. Graphene oxide is used for the purpose.
 - It can also act as gas barrier for e.g. in food packaging.
 - It can be used for separation of organic solvent with water.
 - Biomedical Technologies:** Very **high surface area, electron mobility** etc. is paving the way for novel biomedical technologies. Graphene bioelectronics (transistors and electrode arrays) has become a ground-breaking field that offers existing opportunities for developing new kinds of biosensors. Key

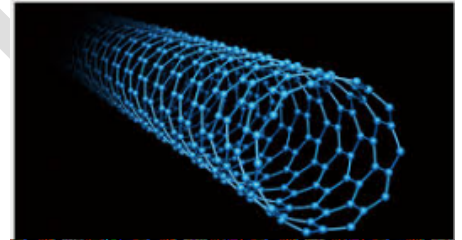
applications include Thermal ablation of highly resistant cancer cells; Bioelectronics (bionics); Electronic interface to living cells and nerve tissues; etc.

- **Sensors:** Since every atom of graphene is exposed it is an ideal material for biological, gas and chemical sensors. It can be used for explosive detection; detecting biomarkers for parkinson's disease; selective gas sensing; self-healable, multifunctional electronic sensor tattoos; environment monitoring etc.
 - Wearable technologies
 - Light weight cars, planes etc.
- **Health Risks:** Extensively debated.
- Toxicity depends on several factors such as shape, size, purity, post-production processing steps, oxidative state etc.

C) CARBON NANOTUBES (CNT)

- Intro

- Carbon nanotubes are allotropes of carbon with a cylindrical nanostructure. These cylindrical carbon molecules have unusual properties, which are valuable for nanotechnology, electronics, optics and other fields of material science and technology.



- Properties

- Strength: One of the most tensile and elastic material discovered yet.

- Wettability

- Exhibits a super hydrophobic property.
- By applying a low voltage as low as 1.3 V, the extreme water repellants surface can be switched into super hydrophilic.

Electrical Properties

- CNT are either metallic or semiconducting along the tubular axis.

Thermal Properties

- All nanotubes are expected to be very good thermal conductors along the tube, exhibiting a property known as "ballistic conduction", but good insulators lateral to the tube axis.

Application

- **Current uses and application** of nanotubes has mostly been limited to the use of bulk nanotubes, which is a mass of rather unorganized fragments of nanotubes.
 - Used as composite fibers in polymers to improve the mechanical, thermal and electrical properties of the bulk product.
 - Tips for atomic force microscope probes
 - In tissue engineering, carbon nanotubes can act as scaffolding for bone growth.

Concerns: Toxicity, health risk not clear yet.

17) ELABORATING ON SOME NANOTECHNOLOGY APPLICATIONS

NANOTECHNOLOGY IN TEXTILE SECTOR

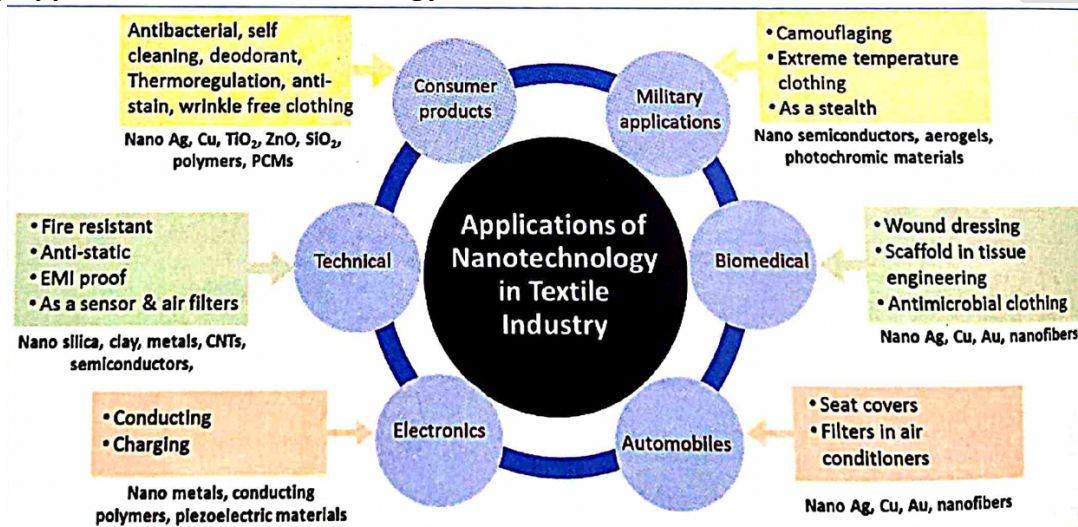
- Example Question

- Discuss the various applications of Nanotechnology in textile sector [10 marks, 150 words]

- Introduction:

- Nanotechnology is an amazing general-purpose technology that has proven its potential in all the fields, and the textile sector is not an exception.

- Key Applications of Nanotechnology in textile sector



- **How antibacterial properties are obtained:** A nano-silver based suspension can be applied on the fabric surface. Other compounds such as zinc, copper, and their oxides coupled with metal nanoparticles can also be used as antibacterial agents. Titanium oxide (TiO₂) is another strong antimicrobial material that conducts the phenomenon known as “Photocatalysis”.
 - The anti-microbial textiles are most essential in the medical sector, especially in the present deadly pandemic conditions, for e.g., in curtains, bed covers, face masks, aprons used in the hospitals and diagnostics center.
 - It can also serve for best hygiene in sportswear, baby products, sanitary napkins, wound healing dressings and so on.
 - **Self-cleaning garments using Titanium oxide particles** save detergent, water, and electricity simply.
- **“Wearable sunscreen”** are garments using Titanium Oxide nanoparticles which provide protection against
- **Thermoregulation:** Clothings respond to changes in the body temperature and maintain comfort by either heating or cooling. The **Phase Change Material** is incorporated into the fabric to fulfill this function.
- The Polymeric micro-capsules filled with perfumes/deodorants/insect repellents can be integrated with fabric for slow release.

- **Fabric Used in Air Filters:** A thin coating of nanofibers on the fabric used in air filters for air conditioners and automobiles' exhaust, stops ultrafine particles thereby increasing the efficiency of the filters.
- **Technical Textiles** are used in special applications. For e.g., fireproof and flame-retardant fabric is required in the suit used by firefighters. The high temperature stable materials are combined with thermally insulating and fireproof materials such as nano-porous silica aerogel for this application. Other examples of technical textiles include functions like electro-magnetic field shielding, electrical conducting, anti-static, camouflaging, stealth, water repellency, and so on. The **nanomaterials used in these applications** are generally carbon nanotubes, nanoparticles of metals, metal oxides, clay, semiconductors, polymers etc.

- Way Forward

- Nanotechnology has a great potential in the textile industry. Functional and smart textiles can play an important role in the economy of a nation. However, there are limitations in applications due to scientific complexity and the cost adhered to it.
- Therefore, a proper study is necessary to find any **toxic effects of handling and wearing the nano materials** to eliminate any health risks due to short- or long-term exposure.

NANOTECHNOLOGY IN HEALTH SECTOR

- **Nanomedicine** is the medical application of nanotechnology. It ranges from the medical applications of nanomaterials and biological devices to nanoelectronics biosensors, and even possible future applications of molecular nanotechnology such as biological machines.
- **What difficulties nano-medicines will overcome?**
 - **Problems with conventional drugs**
 - **Less soluble conventional medicine, which are not absorbed properly and thus don't impact the required target**
 - Even if drug molecule is soluble, body removes the drug before it has had long enough to provide a benefit.
 - Conventional drugs could have side effects due to poor delivery at the actual site of disease.
 - For e.g. drugs targeting cancer must avoid the healthy tissues and organs or damage can be caused.
 - Nanomedicines therefore can play an important role in ensuring enough of the drugs enter the body, that the drugs that enters stays in the body for long periods and is targeted specifically in areas that need treatment.
- **Medical Applications:**
 - **Prevention of disease:**
 - **Nanoparticles of silver** embedded into fibers have anti-microbial action. It is used in food packaging, clothing, disinfectants, and household appliances. Bandages are being infused with silver nanoparticles to heal cuts faster.
 - **Gold Nanoparticles** have anti-bacterial properties

- **Water Purification: Special Filters using nanomaterials** can remove objects as tiny as viruses from water.
- **Diagnostics:** By studying and identifying individual molecules, it is possible to diagnose disease in time to improve the prognosis for the patient.
- **Improved Treatment**
 - Indian Institute of Nano Science and Technology (INST) is developing **Magnetic Hyperthermia mediated cancer therapy** - delivery and localization of magnetic material within the targeted tumour site followed by subsequent application of an alternating Magnetic Field (AMF), thereby generating heat at the tumour site
 - E.g: **Scientist** are using gold nanoparticles to target prostate cancer. Here the nanoparticles or nano shells are made of small payers of Silica glass formed into a sphere and wrapped in a thin layer of gold. This is made to reach the tumour site and then harnessed to cause the tumorous tissue to pulse with extreme temperature when light is applied through a laser specifically designed to excite the particles
 - A team of scientists from IISc Bengaluru have developed nano robots to be used in dental procedure (like root canal therapy).
 - With more advancement in **Nanomachines** – complex surgical procedures would become less intrusive and less complicated

ENVIRONMENTAL NANOTECHNOLOGY

- **Example Questions**
 - What is environmental nanotechnology? Discuss with some practical applications. [10 marks, 150 words]
- **Introduction**
 - Environmental nanotechnology refers to use of nanotech products, processes, and applications to contribute to environment and climate protection.
 - It may contribute by cleaning the pollutants, reducing the pollutants, making our devices more efficient and reducing cost of green technologies.
- **Key areas where nano-material researchers are working:**
 - i. **Ensuring Potable Drinking Water** – Use of Graphene based water filters are expected to increase the accessibility of clean drinking water in coming future.
 - ii. **Removing pollutants from water**
 - **Cleaning up organic chemicals polluting ground water**
 - **Iron nanoparticles** can be used to effectively clean-up organic solvents that are polluting the ground water. The nanoparticles disperse throughout the water and decompose the organic solvents in place.

- **Cleaning up of oil spills**
 - Using photocatalytic copper tungsten oxide nanoparticles to break down oil into bio-degradable compounds.
- iii. **Generating Less pollution during manufacturing of materials**
 - E.g., Use of silver nano particles as catalysts can significantly reduce the polluting by products in the process used to manufacture propylene oxide.
 - Propylene oxide is used to produce common materials such as plastics, paint, detergents and brake fluid.
- iv. **Producing solar cells that generate electricity at competitive cost**
 - E.g. Silicon nanowires embedded in a polymer result in low cost but high efficiency solar cells.
- v. **Increasing the electricity generated by windmills**
 - E.g. use of carbon nanotubes in windmill blades results in stronger and lower weight windmill blades. This helps in more amount of electricity generated by each windmill.
- vi. **Reducing cost of fuel cells**
 - Changing the spacing of platinum atom in fuel cells increases the catalytic ability of the platinum. This allows the fuel cells to function with 80% less platinum, significantly reducing the cost of the fuel cells.
- **Conclusion:** Nanotechnology is thus brimming with potential solutions to various environmental and climate change related issues being faced by India. It's high time that India's Mission on Nano Science and Technology is restructured to include R&D in the field of environment as one of its key goals

18) INDIA AND NANO-TECHNOLOGY

Mission on Nano Science and Technology (Nano Mission)

- Introduction

- a. It is an umbrella program of GoI for overall development in the field of Nanotechnology.
- b. It was launched in 2007 with an allocation of Rs 1,000 crore which was further extended during the 12th five-year plan.
- c. It is structured in a fashion to **achieve synergy** between the national research efforts of various agencies in Nano Science and Technology and launch new programs in a concerted fashion.
- d. Department of Science and Technology is the nodal agency for the mission.

- Objectives of the Nano-Mission

- a. **Basic Research Promotion** – funding of basic research by individual scientists or groups of scientists and creation of centre of excellence for pursuing this research.
- b. **Infrastructure Development** for Nano Science and Technology Research -> development of a chain of facilities across the country.

- c. **Human Resource Development** – Providing effective education and training to researchers and professionals in diversified fields. Launching of M.Sc./M.Tech programmes, create national and overseas post-doctoral fellowships, chairs in universities etc.
- d. **International Collaborations** – Exploratory visits of scientists, organization of joint workshops, conferences and joint research projects, facilitate access to sophisticated research facilities abroad, forge academia-industry partnership.
- e. Development of **product and processes for national development**
 - Especially in areas of national relevance like
 - Safe Drinking Water
 - Materials Development
 - Sensors Development
 - Drug Delivery

- **Achievements of the Mission**

- a. The mission has resulted into more than 5,000 research papers and some useful products like nano-hydrogel based eye drops, pesticide removal technology for drinking water, water filters for arsenic and fluoride removal and nano-silver based on anti-microbial textile coating.
- b. India has moved from the fourth to the third position in the world in terms of scientific publications in nanoscience and technology.

- **Institute of Nanoscience and Technology (INST):** It is an autonomous institute of the Department of Science and technology, Gol.

- **Way Forward:**

- » **Funds**
- » **Human Resource**
- » **Infra – R&D**
- » **Industry-Academia collaboration**
- » **University-University Collaboration**

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Building a solid foundation

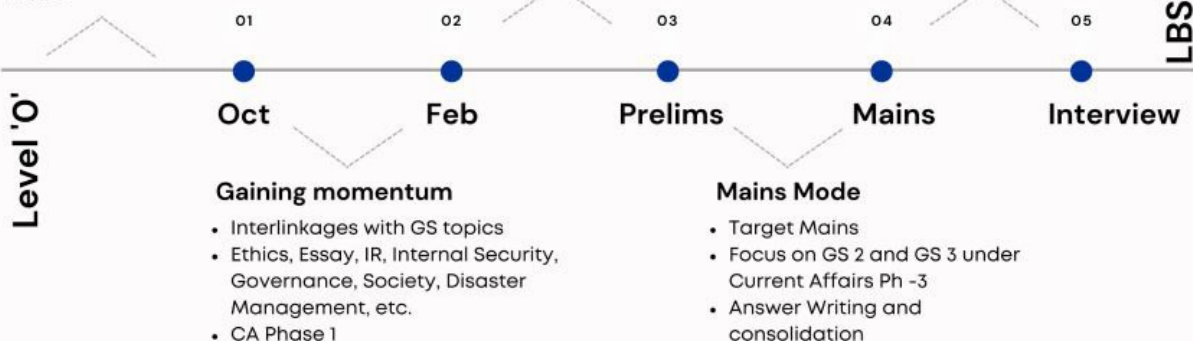
- Focus on fundamentals
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- Current Affairs Phase -1
- Answer Writing structure & basics

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